

FitTrack: project proposal

Oct 9, 2026 · @Maram

FitTrack is a Flutter fitness tracker built by one intern. It covers workout logging, progress charts, goal setting, a BMI calculator and a water intake tracker, with Firebase sign-in and cloud storage.

Overview

People who exercise at home or at the gym lose track of what they did and cannot tell if they are improving. FitTrack keeps workouts, water intake and goals in one place and shows progress as charts.

- **Goal:** a working Android app that one user can sign in to and use every day to log and review their fitness data.
- **Target user:** adults who train two to five times a week and want a simple personal log.
- **In scope:** the eight features in the table below.
- **Out of scope for version 1:** social features, wearable sync, push notifications, diet and calorie tracking, payments.
- **Team:** one developer (individual project).
- **Duration:** four weeks. This is an assumption; adjust it to the internship schedule.
- **Units:** metric (kg, cm, ml).

Features

Six features are required for the first working version; the BMI calculator and water tracker come after them.

#	Feature	What it does	Priority
1	User authentication	Register, log in, log out and reset password with email and password	Must
2	Workout logging	Add, edit and delete a workout: type, duration, calories (optional), notes, date	Must

#	Feature	What it does	Priority
3	Data persistence	Every user's data is saved in Cloud Firestore and stays readable offline through the Firestore cache	Must
4	Progress charts	Weekly and monthly charts of workout minutes and body weight	Must
5	Goal setting	Set a weekly workout-minutes goal, a daily water goal and a target weight, and see progress against each	Must
6	Profile	View and edit name and height; log out	Must
7	BMI calculator	Enter height and weight, get the BMI value and its category, and save the weight as a new entry	Should
8	Water intake tracker	Add water in one tap and see today's total against the daily goal	Should

User flows

A signed-out user always starts at Login; a signed-in user always starts at Home.

First use

1. Open the app and see the Login screen.
2. Tap "Create account".
3. Enter name, email and password, then tap "Register".
4. Land on Home with default goals (150 workout minutes a week, 2,000 ml of water a day).

Log a workout

1. Tap the Workouts tab, then the "+" button.
2. Choose the workout type and date, enter the duration, and optionally calories and notes.
3. Tap "Save".
4. The workout appears at the top of the list and the weekly total on Home updates.

Track water

1. On Home, tap the water card.

2. Tap "+250 ml" for each glass.
3. The ring fills toward the daily goal.

Check BMI

1. On Home, tap "BMI calculator".
2. Enter weight; height is prefilled from the profile.
3. See the BMI value and category, then tap "Save weight" to add it to the weight history.

Review progress

1. Tap the Progress tab.
2. Switch between Week and Month.
3. Read the workout-minutes chart and the weight chart.

Change goals

1. On Home, tap "Goals".
2. Edit the weekly minutes, daily water or target weight.
3. Tap "Save"; Home and Progress use the new goals at once.

Screens

The app has 12 screens. Four of them (Home, Workouts, Progress, Profile) are tabs in the bottom navigation bar.

#	Screen	Purpose	Reached from
1	Splash	Checks the sign-in state and redirects to Login or Home	App start
2	Login	Sign in with email and password	Splash, log out
3	Register	Create an account with name, email and password	Login
4	Forgot password	Send a password reset email	Login
5	Home	Weekly minutes against the goal, water today, last workout, shortcuts to Goals and BMI	Bottom navigation

#	Screen	Purpose	Reached from
6	Workouts	List of workouts, newest first, with a "+" button	Bottom navigation
7	Add / edit workout	Form for type, date, duration, calories and notes	Workouts
8	Progress	Workout-minutes chart and weight chart, by week or month	Bottom navigation
9	Profile	Name, email, height, log out	Bottom navigation
10	Goals	Edit weekly minutes, daily water and target weight	Home
11	BMI calculator	Calculate BMI and save the weight	Home
12	Water tracker	Add water and see today's total against the goal	Home

Backend and API requirements

FitTrack has no custom REST API. The app talks to Firebase through the official Flutter SDKs.

Need	Firebase service	Operations
Accounts	Firebase Authentication (email and password)	Sign up, sign in, sign out, send password reset email, listen to auth state
User profile and goals	Cloud Firestore	Create on register, read as a live stream, update
Workouts	Cloud Firestore	Create, update, delete, stream ordered by date, query by date range
Weight entries	Cloud Firestore	Create, stream ordered by date
Water logs	Cloud Firestore	Read today's document as a live stream, increment the total

Need	Firebase service	Operations
Offline use	Firestore offline cache	Reads and writes work offline and sync when the connection returns

Firebase Storage, Cloud Functions and Cloud Messaging are not used in version 1.

Security rules. A user can read and write only documents under their own user ID:

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    match /users/{userId} {
      allow read, write: if request.auth != null && request.auth.uid ==
userId;
      match /{document=**} {
        allow read, write: if request.auth != null && request.auth.uid ==
userId;
      }
    }
  }
}
```

Database schema

All data lives under one document per user, `users/{uid}` , with three subcollections.

Document path	Field	Type	Notes
<code>users/{uid}</code>	name	string	
	email	string	
	heightCm	number	Optional until the user sets it
	weeklyMinutesGoal	number	Default 150

Document path	Field	Type	Notes
	dailyWaterGoalMI	number	Default 2000
	targetWeightKg	number	Optional
	createdAt	timestamp	
users/{uid}/workouts/{workoutId}	type	string	running, walking, cycling, strength, yoga, other
	durationMin	number	Required, greater than 0
	calories	number	Optional
	notes	string	Optional
	date	timestamp	Day the workout happened
	createdAt	timestamp	
users/{uid}/weightEntries/{entryId}	weightKg	number	
	bmi	number	Calculated from weightKg and heightCm
	date	timestamp	

Document path	Field	Type	Notes
users/{uid}/waterLogs/{yyyy-MM-dd}	totalMI	number	One document per day; the document ID is the date
	updatedAt	timestamp	